

Principled loss functions for continuous-time learning



Mathematical
Institute

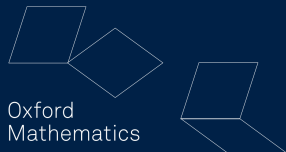
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Motivation

Observed time series are finite sequences of time-stamped values. Sequence to sequence models are trained using pointwise losses such as mean squared error (MSE).

Continuous time models such as neural differential equations enable path-to-path learning, where inputs and outputs are continuous-time trajectories.

An open question is which loss functions are best suited to path-to-path learning, as previous work has shown that pointwise objectives can overfit local noise and fail to capture temporal structure (Ramsay and Silverman, 2005; Kudrat et al., 2025).

Project question

The project studies definitions of path closeness and how each definition changes the learned path or operator. Let $0 = t_0 < \dots < t_m = T$ be observation times and suppose two continuous predictions satisfy

$$\hat{Y}_A(t_j) = \hat{Y}_B(t_j) = Y(t_j), \quad j = 0, \dots, m.$$

Both have zero pointwise mean squared error,

$$\text{MSE}(\hat{Y}_\ell, Y) := \frac{1}{m+1} \sum_{j=0}^m \|\hat{Y}_\ell(t_j) - Y(t_j)\|_2^2 = 0, \quad \ell \in \{A, B\},$$

yet their behaviour between observations can differ. Their elapsed-time errors may therefore differ:

$$J_2(\hat{Y}_A, Y) \text{ need not equal } J_2(\hat{Y}_B, Y), \quad J_2(\hat{Y}, Y) := \int_0^T \|\hat{Y}(t) - Y(t)\|_2^2 dt.$$

Role of discretisation

Although paths are stored at finitely many times, the loss still determines how the underlying paths are compared. Weighting, interpolation and feature representation each affect that comparison.

Functional-data analysis treats an input X and target Y as whole paths (Ramsay and Silverman, 2005):

$$X \in \mathcal{X} := C([0, T]; \mathbb{R}^{d_x}), \quad Y \in \mathcal{Y} := C([0, T]; \mathbb{R}^{d_y}).$$

A parameterized model is an operator from input paths to predicted output paths:

$$\mathcal{F}_\theta : \mathcal{X} \rightarrow \mathcal{Y}, \quad \hat{Y}_\theta := \mathcal{F}_\theta(X), \quad \theta \in \Theta.$$

Given paired training paths $(X^{(i)}, Y^{(i)})_{i=1}^N$ and a path loss $\mathcal{L} : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, \infty)$, training selects

$$\hat{\theta} \in \arg \min_{\theta \in \Theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\mathcal{F}_\theta(X^{(i)}), Y^{(i)}).$$

The loss $\mathcal{L}$ determines which discrepancies between predicted and target paths training treats as important.

Loss	Definition	Emphasis
Pointwise MSE	$\frac{1}{m+1} \sum_{j=0}^m \ \hat{Y}(t_j) - Y(t_j)\ _2^2$	Each recorded point has equal weight
Elapsed-time J_2	$J_2(\hat{Y}, Y) := \int_0^T \ \hat{Y}(t) - Y(t)\ _2^2 dt$	Error accumulated over time
Sobolev H^1	$J_2(\hat{Y}, Y) + \kappa \int_0^T \ \dot{\hat{Y}}(t) - \dot{Y}(t)\ _2^2 dt$	Values and local velocity for smooth paths
Signature loss	$\ \Phi_M(\hat{Y}) - \Phi_M(Y)\ _2^2$	Ordered increments and interactions through depth M

Comparison principle

For the H^1 loss, $Y, \hat{Y} \in H^1([0, T]; \mathbb{R}^{d_Y})$, dots denote weak time derivatives, and $\kappa > 0$ balances value and derivative error (Ramsay and Silverman, 2005). Map Φ_M collects signature coordinates through level M (Chen, 1957; Lyons et al., 2007). Each fitted path is evaluated under every metric.

Following the standard L^p definition (Lieb and Loss, 2001), for residual path $e(t) := \hat{Y}(t) - Y(t)$ and $1 \leq p < \infty$ set

$$\|e\|_{L^p} := \left(\int_0^T \|e(t)\|_2^p dt \right)^{1/p}.$$

If $\tau : [0, T] \rightarrow [0, T]$ preserves elapsed time, then

$$\|e \circ \tau\|_{L^p} = \|e\|_{L^p}.$$

Integral norms therefore record the size and duration of an error, but not the order in which its values occur.

Consequence for rough streams

A second-level perturbation can leave every coordinate value unchanged. Coordinate losses then respond by exactly zero, whatever their exponent or weighting.

Contribution of signatures

Level k is an iterated integral over $t_1 < \dots < t_k$, so it is built from ordering rather than from magnitude at aligned times.

Let $a < b$, let $Y : [a, b] \rightarrow \mathbb{R}^d$ have bounded variation, and let $M \in \mathbb{N}$. For $k = 1, \dots, M$, its signature levels and depth- M truncation are

$$S^{(0)}(Y)_{a,b} := 1, \quad S^{(k)}(Y)_{a,b} := \int_{a < t_1 < \dots < t_k < b} dY_{t_1} \otimes \dots \otimes dY_{t_k},$$

$$S^{\leq M}(Y)_{a,b} := (S^{(0)}(Y)_{a,b}, \dots, S^{(M)}(Y)_{a,b}),$$

where $\otimes$ is the tensor product (Chen, 1957; Lyons et al., 2007). A machine-learning introduction appears in Chevyrev and Kormilitzin (2016).

- ▶ Level 1 records net displacement.
- ▶ Level 2 records ordered pair interactions. Its antisymmetric part records signed area.
- ▶ Higher levels record increasingly detailed ordered interactions.

Truncation and localization

A global comparison uses one truncated signature over the full interval. A local comparison uses signatures over a partition, thereby retaining coarse information about where an ordered feature occurs.

Experiment A: fixed path

Experiment A: fitting one fixed path

Experiment A asks how the training loss changes the path learned from the same finite observations. A Neural ODE is fitted repeatedly to one smooth two-dimensional target, with the target data, model initialization and optimization budget held fixed.

The experiment first compares equal point weighting with elapsed-time weighting. It then asks whether derivative information and signatures recover different aspects of the target, especially a short oscillatory event that is easy to smooth away.

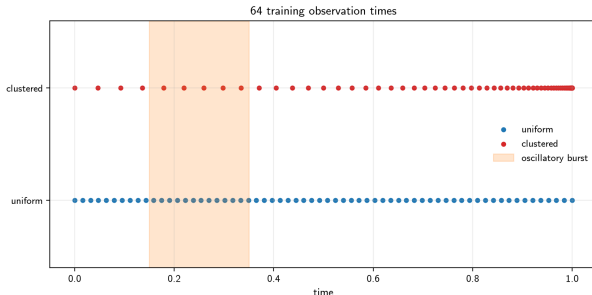
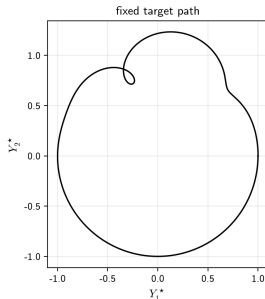
Because there is only one target, this is a controlled path-fitting study rather than an operator-learning task. Its purpose is to expose the geometry and optimization behaviour of the losses before introducing variation across input paths.

Experiment A: target path

For $t \in [0, 1]$, define

$$a(t) := 0.3 \exp\left[-100 \left(t - \frac{1}{4}\right)^2\right], \quad Y^*(t) := \begin{pmatrix} \cos(2\pi t) + a(t) \cos(12\pi t) \\ \sin(2\pi t) + a(t) \sin(12\pi t) \end{pmatrix}.$$

The circular component gives the broad shape. The localized envelope $a(t)$ creates a short oscillatory burst near $t = 0.25$.



A Neural ODE generates a continuous fitted path (Chen et al., 2018):

$$z_{\theta}(0) = z_{0,\theta}, \quad \dot{z}_{\theta}(t) = f_{\theta}(t, z_{\theta}(t)), \quad \hat{Y}_{\theta}(t) = g_{\theta}(z_{\theta}(t)).$$

Here $z_{\theta}(t)$ is the learned hidden trajectory, f_{θ} is a neural vector field, and g_{θ} maps the hidden state to two output coordinates. The restricted model has hidden dimension 2 and width 16. The expressive model has hidden dimension 8 and width 64.

For each loss, the model fits the same target Y^* from 64 observations. The observation times are either uniform or clustered, while the initialization, optimizer and training budget remain matched. Evaluation compares every fitted path on the same dense grid under every metric.

The experiment isolates how the training loss changes the path recovered from the same data and model family.

Experiment A: anchored signature loss

For a two-dimensional output path Y , define the time-augmented path

$$Z_Y(t) := (t, Y(t)) \in \mathbb{R}^{d_Z}, \quad d_Z = 3.$$

The time channel makes the timing of motion visible to the signature. For an interval $I = [a, b]$, write

$$\Delta S_I^{(k)} := S^{(k)}(Z_{\hat{Y}})_{a,b} - S^{(k)}(Z_Y)_{a,b}.$$

Anchor

Signatures do not change when a constant vector is added to every point of a path (Chen, 1957; Lyons et al., 2007). The additional term

$$A_0(\hat{Y}, Y) := \frac{1}{2} \|\hat{Y}(0) - Y(0)\|_2^2$$

restores sensitivity to the starting position. Experiment A treats paths with different initial positions as different, so this information must be retained.

Level weights

Level k has d_Z^k tensor coordinates. Weight d_Z^{-k} converts its squared Euclidean discrepancy into a mean squared discrepancy per coordinate, preventing dimension alone from favouring higher levels.

Experiment A: global and local signature losses

The global objective compares one depth-four signature over the full path:

$$\mathcal{L}_{\text{global}}(\hat{Y}, Y) := A_0(\hat{Y}, Y) + \sum_{k=1}^4 d_Z^{-k} \|\Delta S_{[0,1]}^{(k)}\|_2^2.$$

It measures overall ordered structure, but does not identify where within the interval a discrepancy occurs.

Partition $[0, 1]$ into K intervals $I_b = [(b-1)/K, b/K]$. The local objective is

$$\mathcal{L}_{\text{local},K}(\hat{Y}, Y) := A_0(\hat{Y}, Y) + \frac{1}{K} \sum_{b=1}^K \sum_{k=1}^2 d_Z^{-k} \|\Delta S_{I_b}^{(k)}\|_2^2.$$

Experiment A first uses $K = 10$. Comparing signatures block by block retains coarse temporal location while keeping the representation small.

For the three-dimensional time-augmented path, the two constructions have the same raw size:

$$3 + 3^2 + 3^3 + 3^4 = 120, \quad 10(3 + 3^2) = 120.$$

The comparison changes where structure is retained, rather than simply changing the number of coordinates.

Experiment A: refined local signature loss

On an interval of length h , a level- k signature scales approximately as h^k . Its squared discrepancy therefore scales as h^{2k} . Refining from a reference partition $K_0 = 10$ to $K = 100$ would otherwise make the signature terms much smaller.

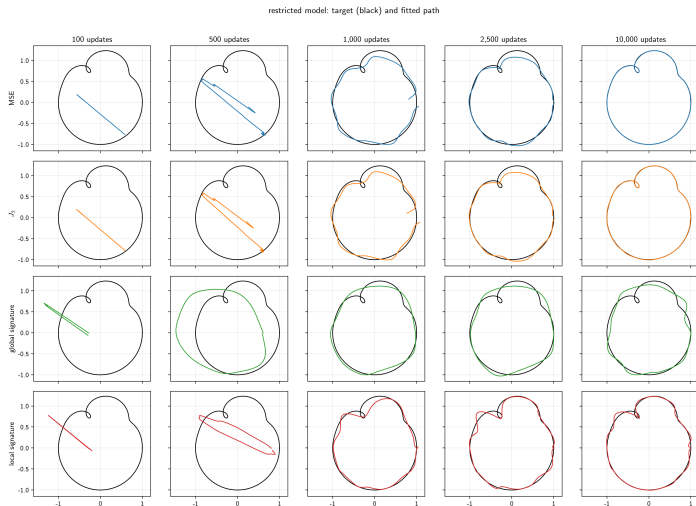
The refined objective compensates for this homogeneity:

$$\mathcal{L}_{\text{local},K}^{\text{hom}}(\hat{Y}, Y) := A_0(\hat{Y}, Y) + \frac{1}{K} \sum_{b=1}^K \sum_{k=1}^2 \left(\frac{K}{K_0} \right)^{2k} d_Z^{-k} \|\Delta S_{I_b}^{(k)}\|_2^2.$$

The factor $(K/K_0)^{2k}$ keeps each level on approximately the same scale as the ten-interval construction. It does not make the fine and coarse losses identical: the finer partition retains more location information.

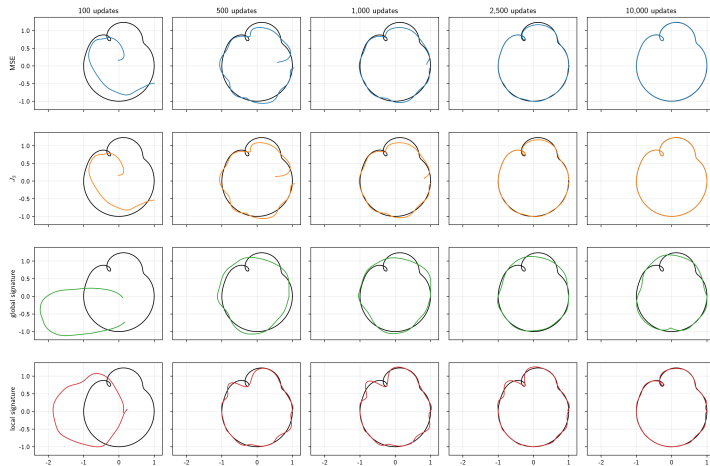
Coordinate averaging and homogeneity correction are project design choices. Their component values and parameter gradients were checked before training.

Experiment A: learning paths with the restricted model

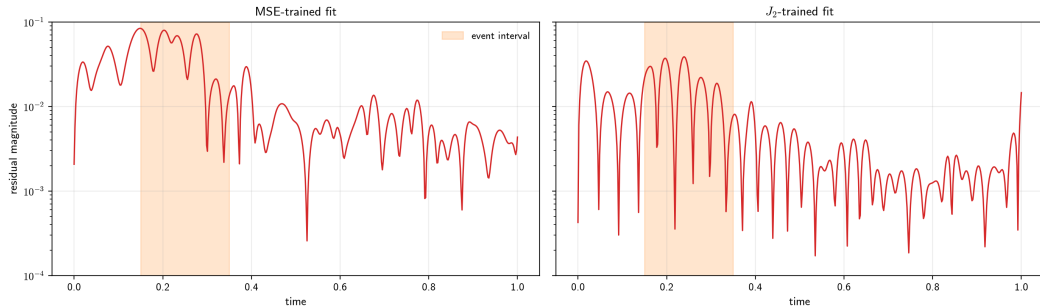


Experiment A: learning paths with the expressive model

expressive model: target (black) and fitted path



Clustered-sampling residuals



One matched restricted-model pair is shown. MSE gives equal weight to every recorded point, so the cluster receives disproportionate influence. The J_2 quadrature weights instead approximate elapsed time and reduce error throughout the path.

Let $E_{c,r}^{\text{MSE}}$ denote dense evaluation MSE for the model trained with MSE at capacity c and seed r . Define $E_{c,r}^{J_2}$ analogously for J_2 training. With $R = 3$ seeds, the reported ratio-of-means reduction is

$$\text{Reduction}_c := 1 - \frac{R^{-1} \sum_{r=1}^R E_{c,r}^{J_2}}{R^{-1} \sum_{r=1}^R E_{c,r}^{\text{MSE}}}.$$

Thus we first average the evaluation error across seeds for each training loss, then compare those two averages. This differs from calculating a percentage reduction within each seed and averaging the percentages.

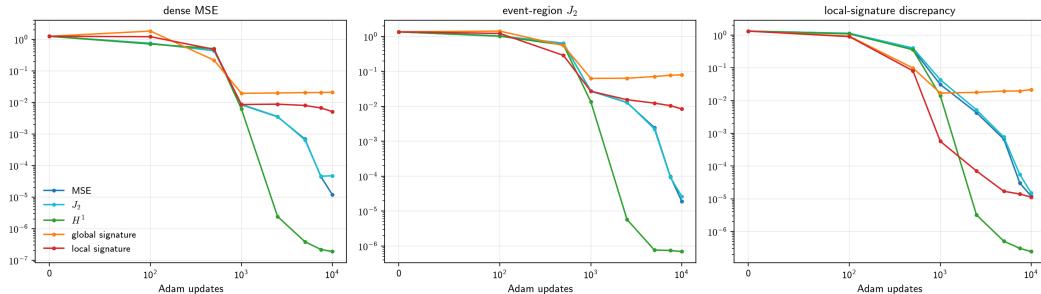
Under clustered sampling, J_2 gives lower dense MSE in all six paired comparisons. The reduction is **68.4%** for the restricted model and **40.8%** for the expressive model. Under uniform sampling, each loss wins three of the six comparisons, as expected when the two objectives differ only through small endpoint-weight effects.

H^1 training gives the best general errors in all six capacity-seed groups at update 1,000 and five of six groups at update 10,000. It directly penalizes velocity error, so it resolves the target's short oscillatory burst efficiently.

This result depends on the smooth fixed target. Its derivative is known analytically, whereas Brownian paths are almost surely outside H^1 . The same derivative supervision is therefore unavailable for rough stream tasks.

Smooth targets can reward derivative-aware losses more than generic structural summaries. Signature losses are compared against this control rather than against MSE alone.

Experiment A: learning dynamics



Common evaluation metrics are shown against Adam updates for the restricted model, seed 0 and uniform observations.

Early objective-specific advantage

At update 500, global-signature training has the lowest global-signature discrepancy in all six capacity–seed groups, while local-signature training has the lowest local-signature discrepancy in five of six. The signature objectives therefore influence what is learned first, even when that influence is not visible in the final fit.

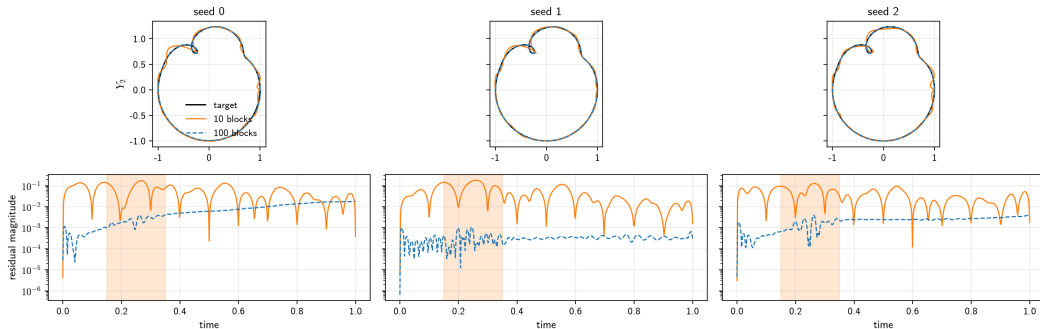
Recovery of the localized event

Neither signature objective leads on error within the oscillatory interval at the same checkpoint. Pointwise and H^1 training are ahead in five of six groups, and H^1 is strongest overall because it directly supervises velocity.

Finite-budget conclusion

Signature advantage is transient and objective-specific. Reporting update 10,000 alone would hide this ordering, while treating the terminal paths as converged optima would overstate what the experiment establishes.

Experiment A: refining the local signature partition



The top row compares fitted paths for 10 and 100 local blocks; the lower row compares their residuals on one shared logarithmic scale. Refining the partition lowers MSE, J_2 , H^1 , maximum error and both local-signature discrepancies in all six capacity–seed groups. Finer blocks preserve the location of the short oscillation more precisely.

Experiment B: learned operator

Experiment B: learning a stream-to-stream map

Experiment B moves from fitting one fixed curve to learning a single rule that acts on many paths. Its input is a complete Brownian driver over $[0, 1]$ and its output is the complete Ornstein–Uhlenbeck response driven by that same stream.

The task is not forecasting: the model produces the response over the same time interval as the observed driver. It must learn how changes in the input stream propagate through the whole output path, then apply that rule to previously unseen Brownian drivers.

This is the project's direct path-to-path learning experiment. It tests whether the sampling-weight effect from Experiment A survives when the learned object is an operator rather than a single fitted path.

Experiment B: target operator and model

For each path i , the target response satisfies

$$dY_t^{(i)} = -2Y_t^{(i)} dt + 0.5 dW_t^{(i)}, \quad Y_0^{(i)} = 0.$$

The same Brownian increments define both $W^{(i)}$ and $Y^{(i)}$, so every pair gives one example of the operator $W \mapsto Y$.

A Neural CDE learns the approximation $\mathcal{F}_\theta : W \mapsto \hat{Y}_\theta$ (Kidger et al., 2020):

$$dh_t^{(i)} = f_\theta(h_t^{(i)}) d(t, W_t^{(i)}), \quad \hat{Y}_\theta^{(i)}(t) = g_\theta(h_t^{(i)}).$$

Here $h^{(i)}$ is a hidden path, f_θ determines how the driver changes that hidden state, and g_θ decodes the predicted response. Separate matched models minimize MSE, J_2 , global-signature loss or local-signature loss.

Experiment B: how the Neural CDE uses a driver

At $t = 0$, the model initializes a hidden state $h_0^{(i)}$ for input stream $W^{(i)}$. As the augmented driver $(t, W_t^{(i)})$ moves, each new driver increment changes the hidden state through the learned vector field f_θ .

The decoder g_θ reads this evolving hidden state to produce a complete predicted response path $\hat{Y}_\theta^{(i)}$. The same parameters θ are used for every training path, so the model learns one rule $\mathcal{F}_\theta : W \mapsto \hat{Y}$, rather than memorizing a separate fitted curve for each input.

During training, $\hat{Y}_\theta^{(i)}$ is compared with the corresponding OU path $Y^{(i)}$ using the chosen path loss. Backpropagation through the CDE solver updates f_θ , g_θ and the initialization map. Changing the loss changes which path discrepancies drive those parameter updates.

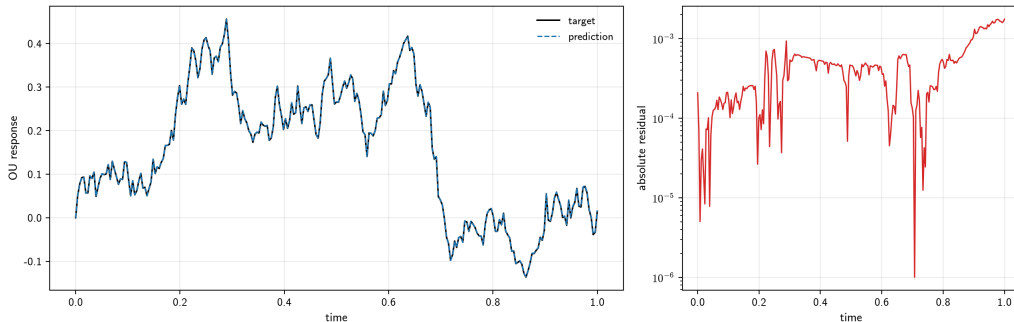
Experiment B: training and held-out comparison

The study contains 512 training, 128 validation and 256 test input–response pairs. Each model sees 64 target times during training; those times are either uniform or deliberately clustered. Within a matched comparison, every training loss receives the same paths, observation times, initialization, minibatch order and 500-epoch Adam budget.

After training, each model predicts every one of the 256 unseen response paths at all 257 time points. The same predictions are scored under dense MSE, elapsed-time J_2 , maximum error, global and local signature discrepancy, and an OU recurrence residual. Thus a model trained with one loss is judged under all common metrics, not only under the objective it minimized.

Results are first averaged over all held-out paths and then over three matched random seeds. A reported percentage such as “51.7% lower” compares these across-seed mean test errors for two training losses.

Experiment B: example held-out response



Target and prediction are shown for one unseen Brownian driver after clustered- J_2 training. The residual panel uses a logarithmic scale; the maximum absolute error is below 2×10^{-3} .

Elapsed-time weighting transfers to operator learning

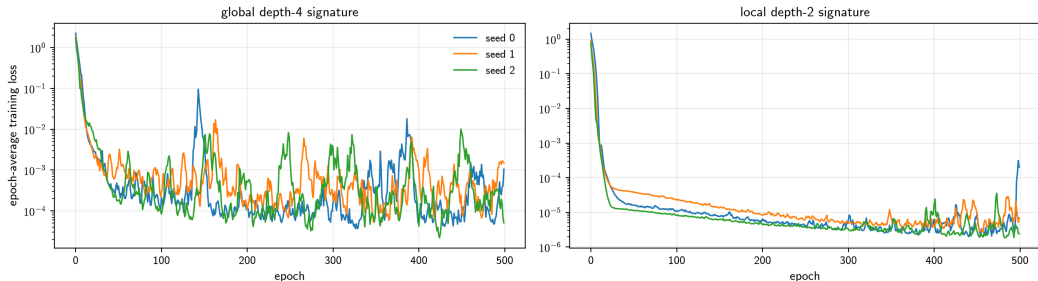
With uniform target times, MSE and J_2 training have essentially tied held-out errors. With clustered target times, J_2 training lowers dense test J_2 in all three paired seeds; its across-seed mean is **51.7% lower** than for MSE training, and dense test MSE falls by 51.6%. The uniform null and clustered separation reproduce the mechanism isolated in Experiment A.

Signature objectives change the learned operator but are less stable

Local-signature training improves its own held-out discrepancy relative to both pointwise fits in two of three seeds, with modest trade-offs in squared error. The remaining local run and all three global-signature runs are unstable at the final epoch, so they do not support a general superiority claim.

The OU operator is deliberately simple and learned to low absolute error. Its role is calibration: it shows that the choice of path loss can matter beyond single-path fitting.

Experiment B: optimizer stability



The left panel shows global-signature training and the right panel local-signature training. Epoch is on the horizontal axis, epoch-average training loss is on the logarithmic vertical axis, and each coloured curve represents one matched seed.

Global-signature loss fluctuates and finishes well above its earlier minimum in every seed: the final-to-best ratios are 28.4, 28.5 and 2.32. Local-signature training is smoother for two seeds but rises late for the third. Final checkpoints therefore reflect optimizer behaviour as well as path geometry, so signature results are reported at a fixed training budget.

Experiment C: sensitivity to messages in Brownian streams

Experiment C: sensitivity to messages in Brownian streams

Experiment C asks how small a deliberately planted feature can be before a path distance stops noticing it. The experiment starts with 10,000 simulated four-dimensional Brownian rough paths and inserts messages of increasing strength into copies of those streams.

No neural network is trained. Each distance is tested directly: first by comparing an original path with its own modified copy, and then by asking whether that distance can separate previously unseen original and modified streams.

The messages can alter visible coordinate motion, hidden second-level area, or both. This creates a controlled setting in which pointwise, integral and signature-based notions of closeness are expected to behave differently.

Experiment C: Brownian rough-path data

Each supplied stream is a sequence of first-level increments and second-level Lévy areas,

$$X = ((a_j, A_j))_{j=0}^{2047}, \quad a_j \in \mathbb{R}^4, \quad A_j \in \Lambda^2 \mathbb{R}^4.$$

Visible coordinate path

Cumulative sums of a_j produce the usual four-dimensional Brownian path. MSE, L^p and maximum error compare only this coordinate motion.

Additional rough-path information

The six area coordinates A_j record within-step ordering that is absent from the coordinate path. An area-only message can therefore leave every visible coordinate unchanged while changing the rough path.

The supplied simulation includes the joint Brownian area information; independent Gaussian coordinate increments alone do not determine it (Wiktorsson, 2001; Foster and Habermann, 2023).

Experiment C: constructing the messages

A message modifies 16 consecutive steps near the middle of a stream. Its normalized amplitude ρ measures the perturbation in units of the natural Brownian standard deviation at the affected level.

Message	Modified component	Intended probe
Increment	First-level increments a_j	Visible coordinate displacement
Area	Second-level areas A_j	Hidden information with no coordinate change
Combined	Both a_j and A_j	Mixed coordinate and area information
Net template	All 16 perturbations have one sign	Accumulated change and endpoint displacement
Balanced template	Eight positive, then eight negative	Local ordering with zero direct sum

The amplitude grid runs from the exact null control $\rho = 0$ to $\rho = 4$. The rough-path composition law is used after modification so that every resulting stream remains algebraically consistent.

Experiment C: distances compared

Path view	Distances	Information available
Coordinate	MSE, J_2 , L^1 , maximum error	Visible coordinate motion
Rough path	Direct area distance	Stored second-level area
Global signature	Variance-normalized depth four	Overall ordered structure
Local signature	Variance-normalized depth two	Ordered structure within short blocks
Response-derived	$U_{r,4}$ for $r \in \{0.5, 1, 2\}$	Levels weighted by a response scale
Truncated kernel	Normalized inner product of depth-four signatures	Similarity in finite signature feature space

Each row defines a different sense in which two streams can be close. All distances are applied to the same original–modified pairs and the same held-out classification split.

Experiment C: variance-normalized signature distance

For block b and signature level k , define

$$\Delta S_b^{(k)}(X, Y) := S_b^{(k)}(X) - S_b^{(k)}(Y).$$

Let q_k^2 be the training variance per level- k coordinate, averaged over unmodified streams and blocks. Since a four-dimensional level- k signature has 4^k coordinates, set

$$d_{\text{var}}(X, Y) := \left[\frac{1}{B} \sum_{b=1}^B \sum_{k=1}^L \frac{\|\Delta S_b^{(k)}(X, Y)\|_2^2}{4^k (q_k^2 + \delta)} \right]^{1/2}, \quad \delta = 10^{-12}.$$

Factor 4^{-k} averages over coordinates, while division by q_k^2 expresses the discrepancy relative to ordinary Brownian variation at that level. The global distance uses one block and $L = 4$; local distances use short blocks and $L = 2$.

Experiment C: response-derived signature distance

In a linear controlled differential equation, a level- k signature coordinate is multiplied by a product of k response matrices. If $\|A_i\|_{\text{op}} \leq r$, then

$$\|A_{i_k} \cdots A_{i_1}\|_{\text{op}} \leq \prod_{j=1}^k \|A_{i_j}\|_{\text{op}} \leq r^k.$$

This motivates

$$U_{r,L}(X, Y) := \frac{1}{B} \sum_{b=1}^B \sum_{k=1}^L r^k \|\Delta S_b^{(k)}(X, Y)\|_1.$$

Small r suppresses higher levels and large r emphasizes them; the fixed family is $r \in \{0.5, 1, 2\}$. The coordinate ℓ^1 norm comes from summing the absolute wordwise contributions in the response bound. This is a project discrepancy motivated by Ferrucci, Perrée and Lyons (2026), not a canonical signature metric.

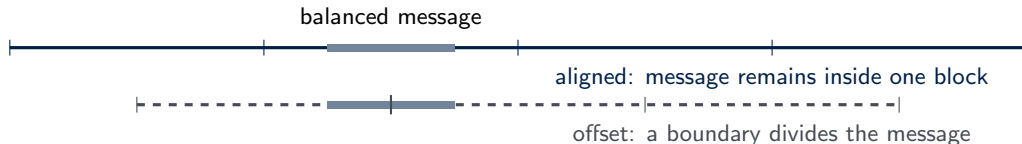
Experiment C: aligned and offset local signatures

Aligned local signature

One comparison block contains the entire balanced message. Its added first-level increments and direct second-level areas sum to zero, so a depth-two signature over that complete block can cancel.

Offset local signature

The same block partition is shifted by eight steps, placing a boundary halfway through the message. Each half is now compared separately, so the cancellation is no longer hidden inside one signature.



Experiment C: normalized paired sensitivity

Let d be one candidate distance and let π pair every unmodified stream with a different unmodified stream. Its natural between-stream scale is

$$c_d := \text{median}_i d\left(X^{(i)}, X^{(\pi(i))}\right).$$

For message type τ , amplitude ρ and template s , define

$$R_d(\tau, \rho, s) := \frac{\text{median}_i d\left(X^{(i)}, \tilde{X}_{\tau, \rho, s}^{(i)}\right)}{c_d}.$$

The numerator measures the planted change while holding the underlying Brownian path fixed. The denominator measures ordinary variation between unrelated Brownian paths. Hence $R_d = 0$ means the distance sees no change, while $R_d = 1$ means the message is as large as a typical between-stream difference under that distance.

Curves of R_d against ρ compare sensitivity across distances with very different numerical scales.

Experiment C: single-stream detection

Paired sensitivity assumes that the original path is available. The population test instead gives the classifier one stream and asks whether it is original or modified.

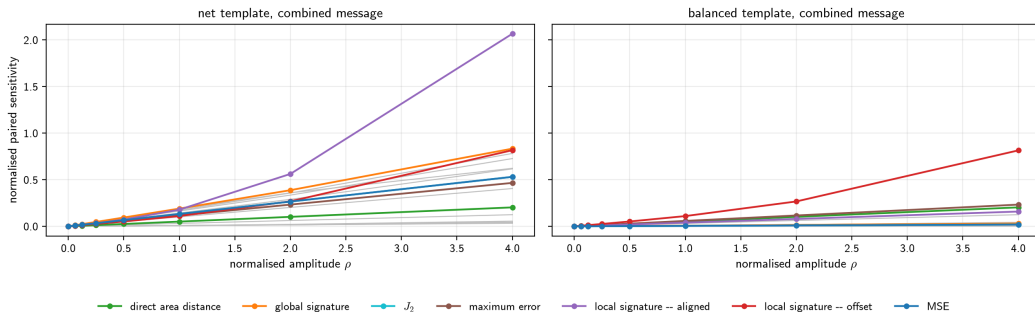
For each distance d , a one-nearest-neighbour classifier assigns the label of the closest training stream:

$$\hat{y}_d(x) = y_{i^*}, \quad i^* \in \arg \min_{i \in \text{train}} d(x, X^{(i)}).$$

It learns no additional feature map, so differences in classification arise from the supplied notion of path distance. Performance is measured by balanced accuracy; chance is $1/2$.

A message is declared detectable at the smallest tested ρ whose 95% lower confidence bound exceeds chance. That amplitude and the preceding grid value are then tested on a separate 10,000-stream ensemble that was not used to choose the threshold.

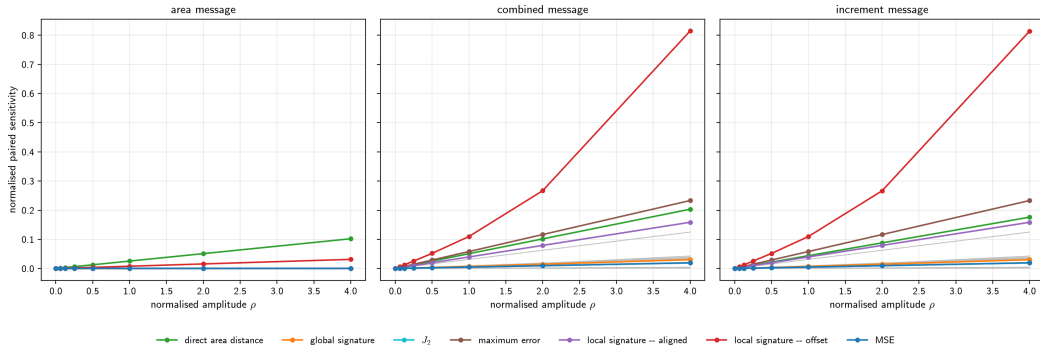
Experiment C: net and balanced templates



Net template. Perturbations accumulate, so endpoint displacement is available and every highlighted distance responds.

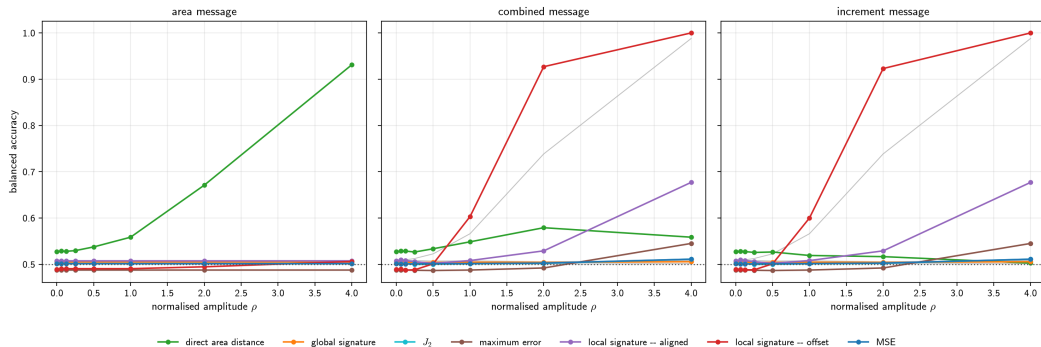
Balanced template. Positive and negative perturbations cancel over the full message block. Sensitivity now depends on whether the representation retains local order and on where its block boundaries lie.

Experiment C: paired response to balanced messages



The three panels differ only in whether the planted message alters area, increments, or both. Higher R_d means that the planted change is larger relative to normal Brownian variation under distance d . Area-only messages are invisible to coordinate losses; combined and increment messages reveal the strong dependence of local signatures on partition placement.

Experiment C: held-out detection results



Each curve uses one named distance; the dotted line is chance balanced accuracy.

Reading the curves

A method becomes reliably informative when its 95% lower confidence bound rises above chance. The operational threshold is the first tested amplitude ρ satisfying that rule, rather than the point at which a curve first appears visually above $1/2$.

Balanced increment and combined messages

Direct increment distance detects from $\rho = 0.5$. The offset local signature detects from $\rho = 1$, the aligned local signature from $\rho = 2$, and maximum error from $\rho = 4$. The different ordering from the paired-sensitivity plots shows that numerical response to a known modification is not the same as population-level detectability.

Independent confirmation

All four thresholds repeat on the second 10,000-stream ensemble. The confirmation ensemble is independent of both classifier fitting and threshold selection.

Conclusions

Evidence	Main result	Meaning of closeness
Sampling limit	MSE targets a density-weighted integral	Observation frequency
Fixed path and OU	J_2 corrects clustered bias	Elapsed time
Smooth fixed path	H^1 resolves a short oscillatory burst	Values and derivatives
Brownian messages	Signature features detect area and order	Ordered multilevel structure

Project conclusion

Two paths count as close only relative to structure relevant to the task. Observation density, regularity, scale, signature truncation, localization and optimization can each change the comparison.

Supported conclusions

Elapsed-time weights correct the emphasis created by irregular observation density. Derivative penalties are powerful when smooth derivatives are meaningful. Signature features are necessary when the relevant distinction is order or area rather than aligned coordinate error, but their truncation, localization and scaling are substantive modelling choices.

Scope

Evidence comes from one fixed target, one OU operator, one supplied Brownian rough-path law, finite optimization budgets and three training seeds. Signature training remains sensitive to optimization, and the Brownian detection thresholds depend on the specified message support, sample size and classifier.

Natural continuation

Hybrid pointwise–signature losses, multiscale local signatures, learned path representations and broader stream-to-stream operators.

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